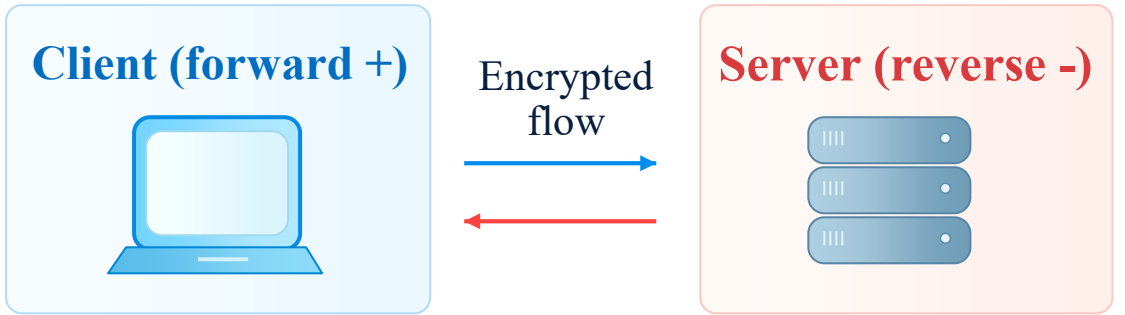
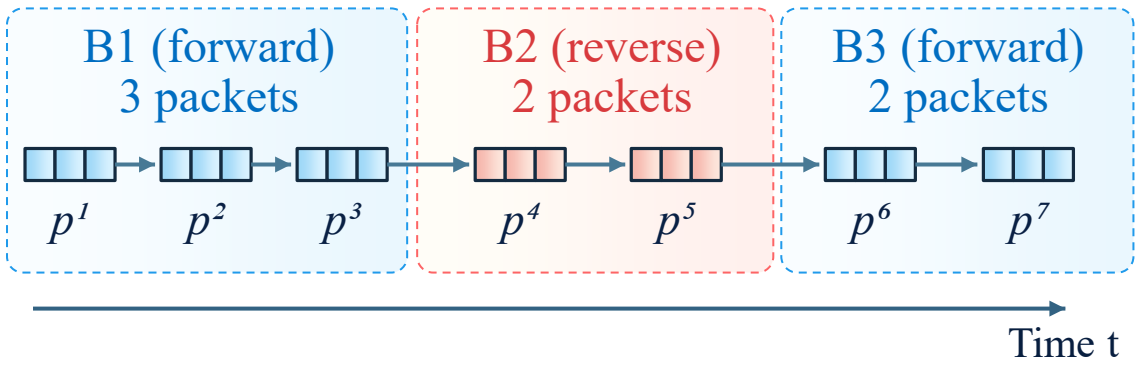
Bidirectional Flow and Bursts

(1) Bidirectional packet sequence



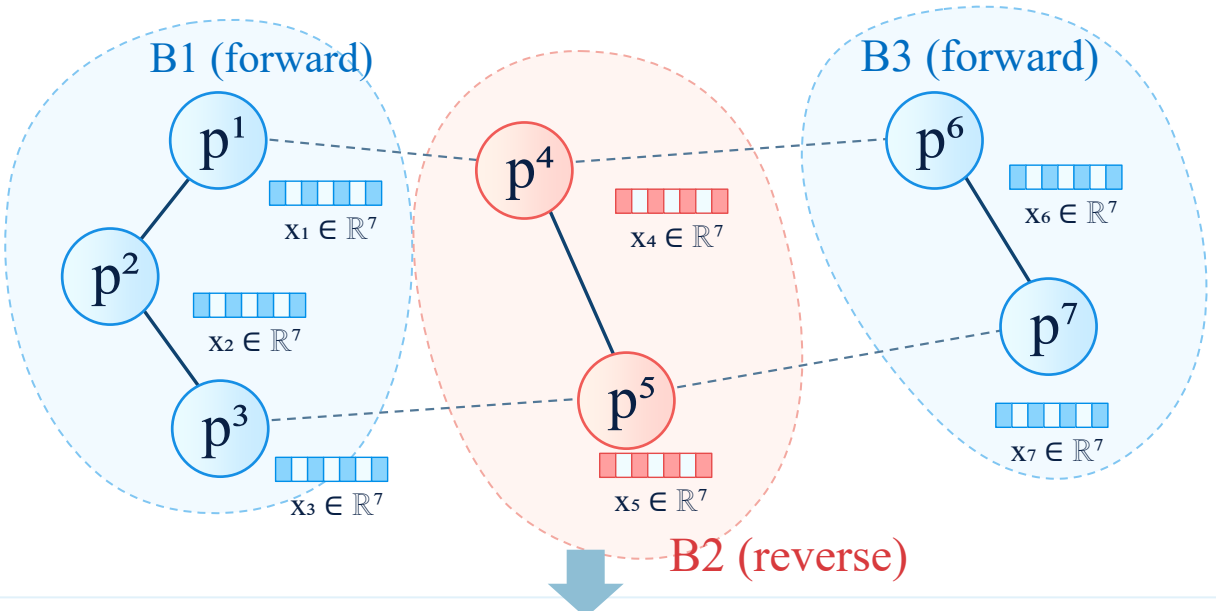
(2) Burst segmentation by direction



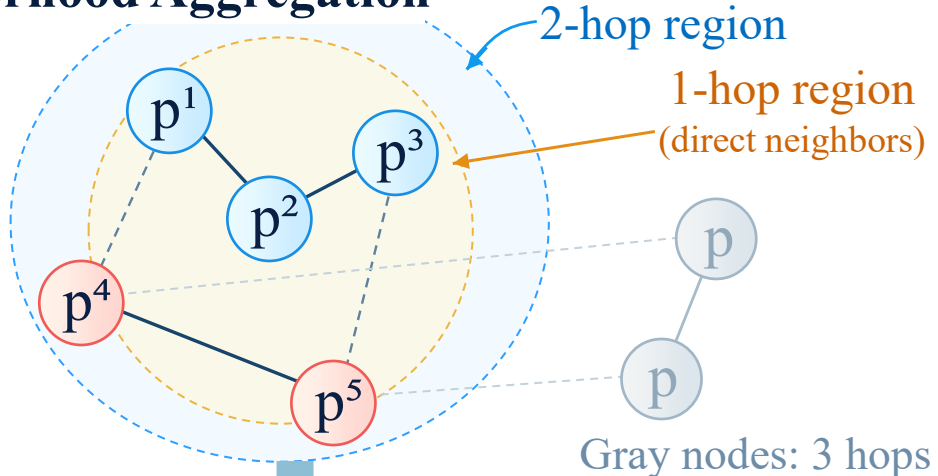
(3) Packet node features

Dir.	Pkt. length	Rel. time	Burst pkts.	Burst bytes	Pkt. ratio	Byte ratio

Flow Interaction Graph G(x) = (V, E)



K = 2 Neighborhood Aggregation



TAGCN encoding (K=2)

$$H = \sigma(\hat{D}^{-1/2}\hat{A}\hat{D}^{-1/2}XW)$$

Global mean pooling

$$z_g(x) = \frac{1}{|V|}\sum_{v \in V} h_v$$

Flow embedding

$$z_g(x) \in \mathbb{R}^{128}$$